BIOMOLECULES AND THEIR APPLICATIPONS

(Module 2 continued notes)

Lipids:

Lipids are a class of biomolecules, insoluble in water and soluble in organic solvents. They are essential component of all living cells. They are sources of energy, form cell membranes and act as signalling molecules.

Types of lipids:

- **Fatty acids**: Building block of lipids, used to build larger molecules such as triglycerides and phospholipids.

- **Triglycerides**: contains three fatty acids attached to glycerol back bone and is the main form of fat stored in the body

- **Phospholipids**: are the components of cell membrane, contains two fatty acids attached to glycerol back bone and a phosphate group

- **Steroids**: are the components of cell membrane, include hormones such as testosterone and estrogen

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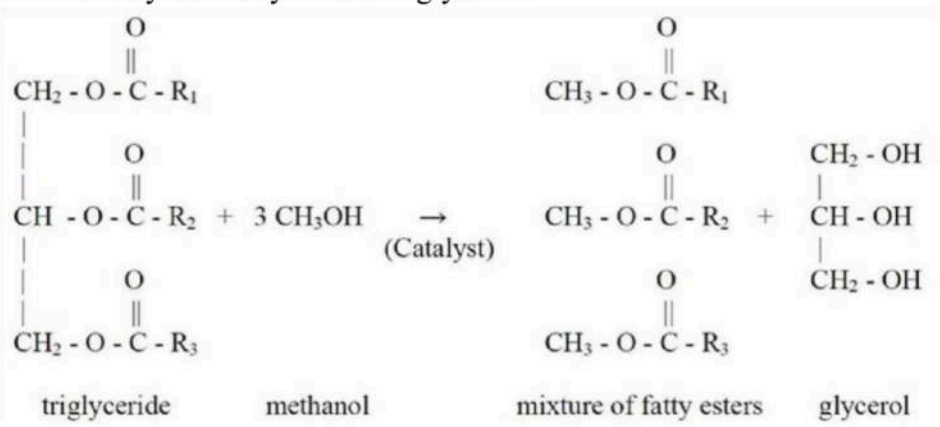
Production of biodiesel from lipids

- **Feedstock preparation:** Vegetable oil, animal fat or recycled cooking oils are taken as a source of lipid. It must be cleaned & dried to remove impurities & water.

- **Trans-esterification:** Feedstock react with alcohol (methanol or ethanol) & a catalyst, such as sodium or potassium hydroxide in a reaction vessel. It breaks down triacylgycerides in the feedstock to fatty acid alkyl esters & glycerol.

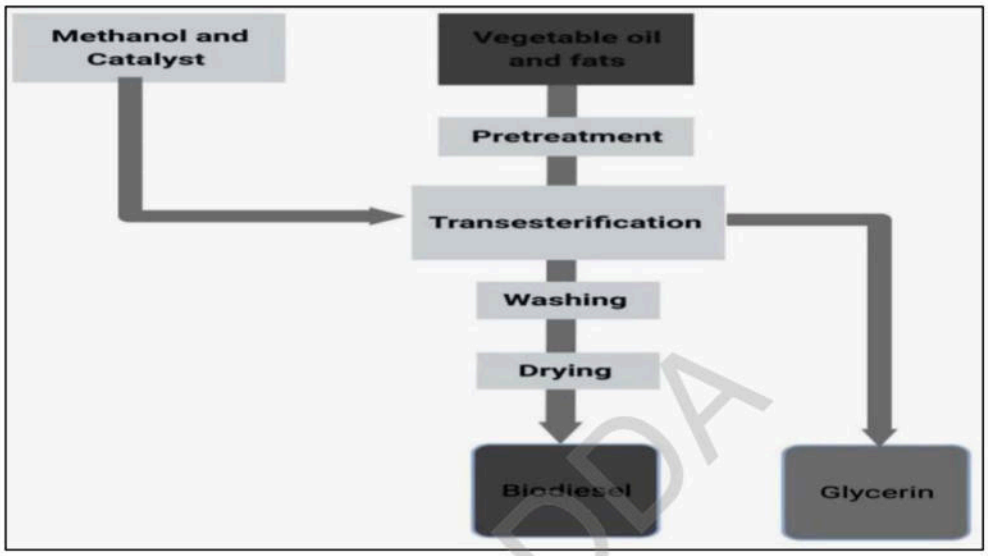


Trans-esterification reaction

- **Separation:** It is carried out using settling tank, heavier glycerol sinks to the bottom & lighter biodiesel floats on top.

- **Washing:** Biodiesel is then washed to remove unwanted catalyst residue or unreacted alcohol.

- **Drying:** To remove remaining water using vacuum or desiccant.



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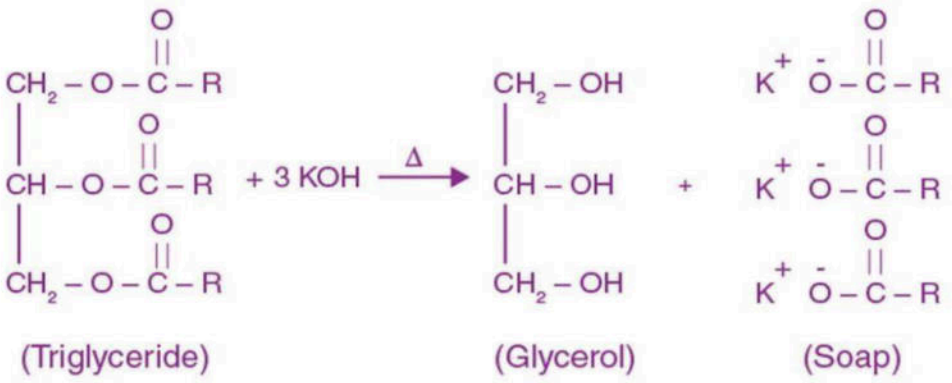
Lipid as detergents:

Mechanism:

- The working principle of lipids as cleaning agents or detergents is based on their ability to dissolve grease and oils.
- Lipids are Act as detergents because of their amphipathic nature, composed of hydrophobic (water-fearing) and hydrophilic (water-loving) regions, which allows them to surround grease and oils, effectively breaking them down into smaller particles that can be more easily removed.
- Because of these properties of lipids, these are commonly used in cleaning products such as soaps, shampoos, laundry detergents, and dishwashing liquids.
- When a lipid-based cleaning agent is applied to a surface, the hydrophobic regions of the lipid molecule surround and dissolve grease and oils, while the hydrophilic regions interact with water, allowing the mixture to be rinsed away.
- The combination of the lipid and water also forms an emulsion, which helps to suspend and remove dirt and debris.
- In addition, some lipids have additional properties, such as foaming or lathering capabilities, that can enhance their cleaning performance.

- For example, fatty alcohols can be used as foaming agents in shampoos, while soap is known for its lathering properties. These additional properties can help to loosen and remove dirt and debris, making the cleaning process more effective.
- Ex: Soap: Saponification is the process in which triglycerides are combined with a strong base to form fatty acid metal salts (soaps).



Saponification reaction

- Eg: Citrus-based cleaning solutions are cleaning products that use natural oil found in citrus fruits such as oranges, lemons, limes, or grapefruits, as their primary active ingredient which is used to break down and dissolve dirt and grime.
- These products can be found in various forms, including sprays, concentrates, and wipes and are available for commercial and household use.

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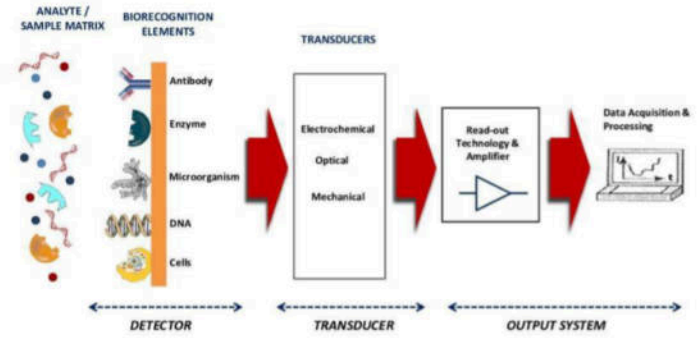
Enzymes:

- Enzymes are proteins that speed up chemical reaction in living organisms that convert substrate to product.
- The sequence of amino acids in enzyme determines the shape of the enzyme & its catalytic activity.

Enzymes in Biosensors fabrication



Schematic representation of the working of different biosensors

- Biosensors are analytical devices that combine a biological recognition element with a transducer to detect and quantify target analytes.
- The biological recognition element can be an enzyme, antibody, nucleic acid, or other biological molecule that specifically interacts with the target analyte.
- The transducer converts the biological response into an electrical signal that can be quantified and interpreted.

Enzymes Used in Biosensors

- **Glucose oxidase**: Used in blood glucose monitoring for people with diabetes. The enzyme oxidizes glucose to gluconic acid and hydrogen peroxide, which is then detected by a transducer to quantify glucose levels in the blood.
- **Lactate oxidase**: Used in the determination of lactate levels in biological fluids, such as blood and urine. Enzyme oxidizes lactate to pyruvate, which is then detected by a transducer.
- **Alkaline phosphatase**: Used in the detection of inorganic phosphates, such as those found in wastewater and fertilizers.
- **Urease**: Used in the detection of urea levels in biological fluids, such as urine. Urease catalyzes the hydrolysis of urea to produce ammonium and carbon dioxide, which can be quantified by a transduce.
- These are just a few examples of the many enzymes that can be used in biosensors to detect and quantify a wide range of target analytes.

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Glucose oxidase in biosensors

Working

- Glucose oxidase is an enzyme that catalyses oxidation of glucose to produce hydrogen peroxide and gluconic acid.

$$\text{D-glucose} + H_2O + O_2 \longrightarrow \text{D-gluconic acid} + H_2O_2$$

- Glucose oxidase is used in diagnostic tests to measure blood glucose levels.
- Biosensor (Glucose amperometric biosensor) consists of test strip that contain small amount of immobilized glucose oxidase, mediators and stabilizers.
- This test strip is inserted into glucose meter, which contains an electrode that measures electric current proportional to amount of glucose in blood.
- The glucose level is displayed on the meter's screen and insulin dose could be adjusted according to the blood glucose level.
- It is an accurate, convenient, and non-invasive procedure.

Advantages of enzymes in biosensors

- ▶ High specificity
- ▶ Enzymatic reaction
- ▶ Wide spread occurrence: Enzymes are sourced from bacteria, fungi
- ▶ Biocompatible
- ▶ Rapid response
- ▶ pH stability
- ▶ Temperature tolerant

Application

- ***Health and biological applications:*** Enzyme-based biosensors play an essential role in the health application by early detection of diseases, pregnancy test kits etc.
- ***Environment and agriculture application:*** Environmental pollutant detection, like detection of pesticides, environment pollution and toxicity before and after bioremediation can be detected.
- ***Bioprocessing industry applications:*** Enzyme-based biosensors can also be used to monitor the presence of products, biomass, antibodies, by-products, enzymes, metabolites, etc., during a fermentation process.
- ***Food processing and drink analysis applications:*** The food and drink industries require a fast and cheaper characterization method to maintain the quality of products and processes.

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Role of enzyme in food processing

- Enzymes acts as catalysts to enhance various processes and improve food quality, texture, and shelf life by breaking down specific components like starch, proteins, and fats.

- **Amylase:** It is used to solubilize the carbohydrates found in barley and other cereals used in brewing.
- **Lipase:** Used to shorten the time for cheese ripening. It is employed in the production of enzyme-modified cheese/butter from cheese curd or butterfat.
- **Papain**: It is widely used as a meat tenderizer.
- **Proteases**: Used in the production of fish meals, meat extracts, texturized proteins, etc.
- **Pectinase**: Treatment of fruit pulp to facilitate juice extraction. It also helps in the clarification and filtration of fruit juice.
- **Lactase**: Additive for dairy products for individuals lacking lactase.

- **Glucose oxidase:** Conversion of glucose to gluconic acid to prevent Maillard reaction (reaction that gives browned food a particular flavors) in products caused by high heat used in dehydration.
- **Cellulase:** Conversion of cellulose waste to fermentable feedstock for ethanol or single-cell protein production.

Benefits of Using Enzymes in Food Processing:

- **Improved Food Quality:** Enzymes can enhance texture, flavor, and aroma in various food products.
- **Enhanced Processing Efficiency:** Enzymes can speed up processes, reduce processing time, and increase yields.
- **Improved Shelf Life:** Some enzymes can help preserve food by inhibiting microbial growth or preventing enzymatic reactions that lead to spoilage.
- **Better Nutritional Value:** Enzymes can improve the digestibility and bioavailability of nutrients in food.

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Enzymes in detergent formulation

- Enzymes are added to detergent formulations to enhance stain removal, making detergents more effective and environmentally friendly by breaking down tough stains like proteins, fats, and carbohydrates.
- Enzymes function by breaking down specific types of stains into smaller, water-soluble molecules that can be easily washed away.
- Different enzymes target different stain types, enhancing the cleaning power of detergents.

- **Proteases** – Break down protein-based stains (blood, egg, grass, sweat)
- **Amylases** – Remove starch stains like gravy, sauces.
- **Lipases** – Dissolve fat and grease stains (butter, oil)
- **Cellulases** – Restore fabric brightness by breaking down tiny cotton fiber fuzz
- **Pectinases** – Remove plant-based stains from sauces, fruits, and coffee.